



# IMMUNE EPITOPE DATABASE AND ANALYSIS RESOURCE

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# Immune Epitope Database

The Immune Epitope Database and Analysis Resource (IEDB) contains information related to antibodies and T cells across an expansive scope of research fields (infectious diseases, allergy, autoimmunity, and transplantation).

The Immune Epitope Database (IEDB) captures experiments that identify and characterize epitopes and epitope specific immune receptors along with various other details such as host organism, immune exposures, and induced immune responses. A companion site, IEDB-Analysis Resource (IEDB-AR), hosts various B and T cell epitope prediction tools based on algorithms trained and validated on the IEDB data along with epitope analysis tools

New epitope prediction and analysis tools are regularly added in the IEDB-AR with features to advance epitope-based therapeutics and vaccine development. For example, a tool to reduce undesired immunogenicity of therapeutic proteins was implemented recently

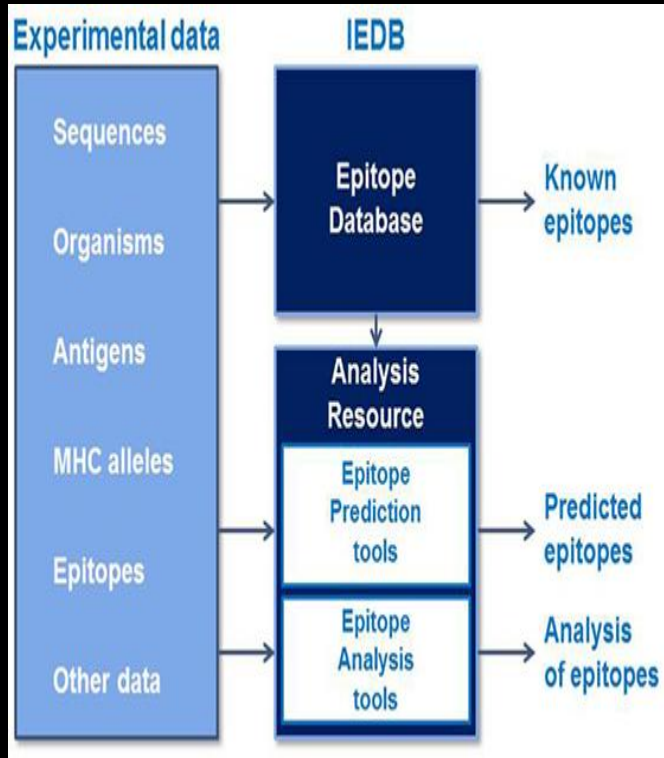
The screenshot displays the IEDB-AR website interface, which is organized into several functional sections:

- Welcome:** A text block explaining that the IEDB is a free resource funded by a contract from the National Institute of Allergy and Infectious Diseases, offering easy searching of experimental data.
- Summary Metrics:** A table providing an overview of the database's content:

| Metric                   | Count   |
|--------------------------|---------|
| Peptidic Epitopes        | 431,456 |
| Non-Peptidic Epitopes    | 2,616   |
| T Cell Assays            | 333,518 |
| B Cell Assays            | 406,439 |
| MHC Ligand Assays        | 891,549 |
| Epitope Source Organisms | 3,648   |
| Restricting MHC Alleles  | 760     |
| References               | 19,208  |
- START YOUR SEARCH HERE:** A central area with multiple search filters:
  - Epitope:** Radio buttons for Any Epitopes, Linear Epitope, and Non-peptidic Epitopes. Includes an "Exact M" field with an example "SINPEKL".
  - Assay:** Checkboxes for Positive Assays Only, T Cell Assays, B Cell Assays, and MHC Ligand Assays. Includes an "Ex: neutralization" field.
  - Antigen:** A field for Organism (e.g., "influenza, peanut") and Antigen Name (e.g., "core, capsid, myosin").
  - MHC Restriction:** Radio buttons for Any MHC Restriction, MHC Class I, MHC Class II, and MHC Nonclassical. Includes an "Ex: HLA-A\*02:01" field.
  - Host:** Radio buttons for Any Host, Humans, Mice, and Non-human Primates. Includes an "Ex: dog, camel" field.
  - Disease:** Radio buttons for Any Disease, Infectious Disease, Allergic Disease, and Autoimmune Disease. Includes an "Ex: asthma, diabetes" field.
- Epitope Analysis Resource:** A sidebar containing tools for T Cell Epitope Prediction, B Cell Epitope Prediction, and Epitope Analysis Tools.

At the bottom, there are links for "Create Feedback", "Data Release", "Database Contact", and "Data License Information". A footer note states: "Supported by a contract from the National Institute of Allergy and Infectious Diseases, a component of the National Institutes of Health in the Department of Health and Human Services. Data Last Updated: March 25, 2018."

# Which data are included in IEDB



The database also includes nonpeptidic epitopes. These include carbohydrates, lipids, chemicals, metals, and drugs. In addition, for all data there is a minimal set of information required. For linear peptides, the amino acid sequence must be specified, and for discontinuous epitopes, the amino acid and their respective position must be specified (e.g., G104, G106, L107, and W231)

The IEDB does not capture predicted/inferred data or reviews. All data come from empirically derived assays reflecting the interaction of an adaptive immune receptor with an epitope. There are four categories of assays—T cell, B cell (including soluble antibodies), MHC binding, and MHC ligand elution.

The goal of the IEDB is to allow a global query of all data from the scientific literature and direct submission. Individual epitopes can be tested in multiple references, in more than one host and under different conditions. If an epitope has been studied in multiple publications, users can gather all the cumulative data from the IEDB and draw their own conclusions regarding its utility for their application.

# Direct submission process



## Data Submission Overview

### Submission:

Submitter submits files via IEDB data submission tool

### IEDB review:

Curator reviews data

Communication with submitter if necessary to modify submission

Submitter sent proof of submission for approval

### Submitter Review:

Submitter reviews and approves proof for public release, requests modifications, or places data on hold from public release

- The IEDB is populated with data from direct submissions in addition to literature curation. Direct submissions account for about 20% of the epitope records. The primary source of submissions has been the NIAID-funded large-scale T cell and antibody epitope discovery contracts and grants for infectious diseases, such as dengue virus, influenza virus, arenavirus, tuberculosis, and common allergens, although other researchers have deposited their epitopes as well.
- There are three ways to submit data to the IEDB. The first is a wizard system that walks the submitter through a step-by-step process to enter data. This is ideal for new submitters who need to become familiar with the data fields and for those with only a few epitopes and assays. The second method utilizes Excel files that work as templates for different types of assays. This method accommodates larger data submissions and is the most commonly used. The third method uses XML files, which offer the greatest flexibility, but also require expertise in the use of XML and familiarity with the IEDB data structure. One of the IEDB curators specializes in assisting researchers in submitting data.



# IEDB basic search

The IEDB provides users with two basic options—the home page search and specialized searches. In both cases, special “finder” features assist in this process. Data field names are explicit and concise by necessity and may be unfamiliar to some users. For example, it is necessary to distinguish whether a protein is delivered as a vaccine or if it is used on an ELISA plate. The protein used to immunize is called the immunogen, and the protein on the plate is called the antigen. The source of peptides is the epitope’s source antigen. The search interface on the IEDB home page allows performing the majority of user queries and provides the ability to refine initial queries on the subsequent results page. The home page interface contains six sections or panes that allow users to search by major categories of epitope, antigen, host, assay, MHC restriction, and disease. Clicking the green Search button brings the user to the results page. This page contains panels on the left-hand side that allow for further filtering of the results using the same six elements as before, plus a pane for references. Although some of these panes are similar to those on the home page, they allow users to be more specific in selecting search criteria and thus refining the query.

**A**

**B Cell Search**

Reference

Epitope

Immunization

Host Organism

Organism Finder

Host Details

1st In Vivo Process

1st Immunogen

Epitope Relation

Type

Organism

Immunogen Details

2nd In Vivo Process

2nd Immunogen

Immunization Comments

Adoptive Transfer

B Cell Assay

Qualitative Measurement

Assay Type

Assay Finder

Measurement Details

Assayed Antibody

Antigen

3D Structure of Complex

Complex PDB ID

Antibody Chain 1 PDB Chain

Antibody Chain 2 PDB Chain

Antigen PDB Chain

Viewer Flag

Comments

Curated Contacts

Calculated Contacts

Epitope Residues

Antibody Residues Interacting with Antigen

**B**

Search Parameters:

- 1st In Vivo Process is present
- 1st Immunogen is present
- In Vivo1 Immunogen Epitope Relation equals 'Source Organism'
- In Vivo1 Immunogen Object Subtype equals 'Organism'
- In Vivo1 Immunogen Object Organism Name contains 'Toxoplasma gondii'
- Complex Calculated Antibody Antigen Residues contains 'W'

Revise Search

| Epitopes     | Positive* | Negative** | All |
|--------------|-----------|------------|-----|
| Peptidic     | 1         | 0          | 1   |
| Non-Peptidic | 0         | 0          | 0   |

| Assays             | Positive | Negative | All |
|--------------------|----------|----------|-----|
| T Cell Response    | 0        | 0        | 0   |
| B Cell Response    | 1        | 0        | 1   |
| MHC Ligand Elution | 0        | 0        | 0   |
| MHC Binding        | 0        | 0        | 0   |

**C**

IMMUNE EPITOPE DATABASE AND ANALYSIS RESOURCE

Home Browse Search Tools Support

Epitope

1 item(s) found, displaying 1 to 1 (Click the column headers to adjust the sorting)

Export all results: (full)

| Epitope ID | Structure                                             | Source Antigen                      | Source Organism   |
|------------|-------------------------------------------------------|-------------------------------------|-------------------|
| 77491      | T85, A86, L87, E89, P91, T92, Y95, N98, Q100, S109... | Major surface antigen p30 precursor | Toxoplasma gondii |

1 item(s) found, displaying 1 to 1

Export all results: (full)

**D**

IMMUNE EPITOPE DATABASE AND ANALYSIS RESOURCE

Home Browse Search Tools Support More IEDB

Keyword Search Search

Epitope Information

Distinct Epitope

Epitope ID: 77491

Discontinuous Residues: T85, A86, L87, E89, P91, T92, Y95, N98, Q100, S109, C110, T111, S112, K113, A114, V115, I162, K163, G164

Source Antigen: Major surface antigen p30 precursor

Source Organism: Toxoplasma gondii

Source (1) Reference (1) B Cell Assay (2)

2 item(s) found, displaying 1 to 2 (Click the column headers to adjust the sorting)

Export all results: (compact | full)

| B Cell ID | Reference                    | Epitope                                                                                                     | Host         | Immunization                                                    | Assay Antigen                                                                             | Antigen Epitope Relation | Assay Description                                                |
|-----------|------------------------------|-------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------|
| 1245875   | Marc Gralle; J Mol Biol 2005 | T85, A86, L87, E89, P91, T92, Y95, N98, Q100, S109... Major surface antigen p30 precursor Toxoplasma gondii | Mus musculus | Administration in vivo with Toxoplasma gondii (Source Organism) | Toxoplasma gondii Toxoplasma gondii                                                       | Source Organism          | Ab Competition of Ab binding Detection of Ab/Ig binding Positive |
| 1245879   | Marc Gralle; J Mol Biol 2005 | T85, A86, L87, E89, P91, T92, Y95, N98, Q100, S109... Major surface antigen p30 precursor Toxoplasma gondii | Mus musculus | Administration in vivo with Toxoplasma gondii (Source Organism) | Major surface antigen p30 precursor Major surface antigen p30 precursor Toxoplasma gondii | Source Antigen           | X-Ray Crystallography Characterization of Ab binding Positive    |

2 item(s) found, displaying 1 to 2

Export all results: (compact | full)

# Specialized search options

The screenshot shows the 'MHC Restriction' search interface. It features a 'Find' button and a 'Resolution' dropdown menu set to 'Any'. Below the resolution menu is a 'Select multiple options' dropdown with 'Select All' and 'Unselect All' buttons. The 'Evidence' section includes a 'Host' dropdown menu set to 'Any' and a list of checkboxes for various search criteria: 'Single allele present', 'T cell assay -MHC subset identification', 'Elution with MHC specific antibody', 'T cell assay -Mismatched MHC molecules', and 'MHC binding assay'.

The specialized searches can be accessed from the home page under the Specialized Searches pull-down menu. The first four items in the list, Epitope Details, T Cell Assay Details, B Cell Assay Details, and MHC Assay Details, allow the user to access every field in the database, greatly expanding the search options.

The final specialized search is the Identifier Search, which is divided into two sections. The top section has query fields for internal IEDB identifiers for Epitope ID, Reference ID, Submission ID, and Assay ID. All curated articles and submission are given a Reference ID. In addition, data submissions are given a Submission ID that researchers can use in their publications. When the paper is published and curated, a link in the database is created between the paper and corresponding submission. All assays and epitopes are also assigned unique identifiers. The bottom section of the Identifier Search contains query fields for identifiers assigned by three external databases that have links to the IEDB.

The first Specialized Search in the menu is Epitope Details. This page has a similar layout to the Results page in the home page search interface. It has the same four results tabs, but the filter selections on the left are different and can be expanded to reveal numerous fields that were not previously available. The Epitope Reference Details section contains fields for epitope name and start/stop positions in a reference protein. The Epitope Related Object section includes fields intended to capture entities that bear some relationship to the epitope—**analog, mimotope, and neoepitope**. **Analog**s are synthetic constructs of peptide sequences or chemical compounds that share some structural features in common with another sequence or compound. They are often used to determine the role of specific amino acids in the binding or immunogenicity of an epitope. The source of an analog is always artificial. **Mimotopes** are functional mimics of natural molecular structures which bear little or no sequence homology to their biological counterparts. **Cancer neoepitopes** are defined herein as any epitope comprising amino acids that are not encoded in the germ line genome but that arise due to somatic (nonsynonomous) mutations, excluding somatic rearrangements of TCR and Ig genes. The IEDB will curate both immunogenic neoepitopes and putative epitopes tested for recognition that were **not found to be immunogenic**.

# An example query using home page interface



The screenshot displays the MHC Peptide Binding Prediction (MPEP) web interface. The interface is organized into three main sections: Epitope Structure, Epitope Source, and Immune Recognition Context.

**Epitope Structure:**

- ☐ Any
- ☒ Linear Peptide:
- ☐ Discontinuous Peptide
- ☐ Non-Peptidic:

**Epitope Source:**

- Source Organism:
- Source Antigen:

**Immune Recognition Context:**

- ☐ B Cell Response
- ☒ T Cell Response
- ☐ MHC Binding
- ☐ MHC Ligand Elution

Host Organism:

MHC Restriction:

MHC Class:  ▾

We use as an example the well-characterized epitope from influenza A nucleoprotein (NP) protein, ASNENMETM. This can be done by entering or pasting the sequence in the Epitope pane sequence text field on the home page and clicking the Search button. As of January 2017, performing this query results in five epitopes, including the naturally occurring sequence from influenza A virus, and four entries representing that epitope with different posttranslational modifications. One of them, ASNENMETM + MCM(E7), indicates the main chain modification of residue seven. Another, ASNENMETM + GLY(E4), indicates a glycosylation of residue four.

The Assay tab itself has three tabs for T cell, B cell, and MHC Ligand assays (binding and elution). Most of the assays from this query are T cell , with the others being MHC ligand assays (50). Each row in the Assays tab result table represents the elements for one assay. The columns indicate the reference, epitope, host, immunization process, the assayed antigen, its relationship to the epitope, the MHC restriction (if defined and recorded), the assay type, and the qualitative value (positive or negative). By clicking on the Assay ID in the first column, users can access the relevant Assay Details page, in order to see the full curation. The Reference tab provides a summary of the articles and submissions that were curated.



Current Filters: ✖ Positive Assays Only ✖ Epitope Structure: Linear Sequence ✖ Linear Sequence: ASNENMETM

| Epitopes<br>(5) | Antigens<br>(1) | Assays<br>(375) | References<br>(135) |
|-----------------|-----------------|-----------------|---------------------|
|-----------------|-----------------|-----------------|---------------------|

Go To Records Starting At  GO Export Epitopes Results ✖

5 Records Found ⏪ ⏩ Page 1 of 1 ⏪ ⏩ 25 ▾ Per Page

| Details ▾ | Epitope ▾                           | Antigen ▾                    | Organism ▾                       | # References ▾     | # Assays ▾ |
|-----------|-------------------------------------|------------------------------|----------------------------------|--------------------|------------|
| 4602      | ASNENMETM <span>▾</span>            | Nucleoprotein <span>▾</span> | Influenza A virus <span>▾</span> | 134 <span>▾</span> | 371        |
| 4600      | ASNENMETM + MCM(E7) <span>▾</span>  |                              |                                  | 1                  | 1          |
| 4601      | ASNENMETM + MCM(M6) <span>▾</span>  |                              |                                  | 1                  | 1          |
| 161084    | ASNENMETM + GLYC(E4) <span>▾</span> | Nucleoprotein <span>▾</span> | Influenza A virus <span>▾</span> | 1 <span>▾</span>   | 1          |
| 161085    | ASNENMETM + GLYC(M6) <span>▾</span> | Nucleoprotein <span>▾</span> | Influenza A virus <span>▾</span> | 1 <span>▾</span>   | 1          |

- On the results page, users can take advantage of the various filter functions embedded throughout the tabs. For example, each epitope record in the Epitopes tab has a funnel icon to the right of the sequence. Clicking on this icon redisplayes the query results to show data only for that epitope. In the case of the epitope ASNENMETM with Epitope ID 4602, clicking on the filter icon generates results with only one epitope, one antigen, 371 assays, and 134 references, slightly less data than the original query since the posttranslational modifications are omitted. The Assay tab itself has three tabs for T cell, B cell, and MHC Ligand assays (binding and elution). Most of the assays from this query are T cell, with the others being MHC ligand assays. Each row in the Assays tab result table represents the elements for one assay. The columns indicate the reference, epitope, host, immunization process, the assayed antigen, its relationship to the epitope, the MHC restriction (if defined and recorded), the assay type, and the qualitative value (positive or negative). By clicking on the Assay ID in the first column, users can access the relevant Assay Details page, in order to see the full curation.



# Example queries for human diabetes epitopes

- The first query seeks to specifically identify all human data available for type 1 diabetes (T1D), including what antigens have been mapped for T cell epitopes, and how many therapeutic epitopes have been identified. The query can be performed with the home page search by clicking Search to get to the Results page and then perform the following actions: in the Disease pane, type “diabetes” in the Specific Disease text field and select “diabetes mellitus” from the auto-complete list. In the Host pane, select the “human” radio button and in the Assay pane, uncheck B Cell Assays and MHC Ligand Assays, and click the Search button. The results of the search show over 450 epitopes. The Antigens tab displays the antigens in descending order of the number of epitopes. In this case, Glutamate decarboxylase 2 (GAD), one of the predominant antigens for type 1 diabetes, appears at the top of the list.
- Next, to find the subset of these data that is associated with treatments or decreased disease, the user can refine the initial search by adding the criteria of T cell assays related to treatment. In the T Cell Assay text field, type “treatment” and select “decreased disease in vivo assay (reduction of disease after treatment)” and click the Search button. As of January 2017, this search revealed two epitopes found to be therapeutic in humans with T1D. The final example query builds on the previous one, asking if there are immunodominant regions defined for insulin in humans with T1D. Perform the query as above for humans, T1D and T cell responses. Then, on the antigen tab, click the Immunome Browser icon next to insulin. This will display the response frequency of all assays mapped onto a reference antigen for insulin. From this, the user can visualize the immunological hot spots.

IMMUNE EPITOPE DATABASE AND ANALYSIS RESOURCE

Home Browse Search

Search

Epitope Structure

☒ Any ☐ Linear Peptide ☐ Discontinuous Peptide ☐ Non-Peptide

Exact Match Molecule Find

B Cell Search T Cell Search MHC Binding Search MHC Ligand Elution Search Epitope Search Identifier Search

## Select T cell Search

Epitope

Epitope ID

Type ☒ Any ☐ Linear peptide ☐ Discontinuous peptide

Source Molecule Insulin;Insulin precursor;Chain B, Structure Molecule Finder

Source Organism Organism Finder

Epitope Reference Details

Epitope Related Object

## Choose human insulin as epitope source

T Cell Assay

Qualitative Measurement ☒ Positive ☐ Positive-Low

Assay IFNγ release by ELISP

## Specify IFNγ ELIS

| Epitopes    | Peptides |
|-------------|----------|
| Peptide     |          |
| Non-Peptide |          |

Assays

☒ T Cell Responses ☐ B Cell Responses ☐ MHC Ligand Elution ☐ MHC Binding

Summary

## Results

# IEDB prediction of MHC I and MHC II

## IEDB Analysis Resource

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### MHC-II Binding Prediction Results

Input Sequences

| # | Name                                  | Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | West Nile virus envelope glycoprotein | FNCLGHSIRDFLEGVSGATWDLVLEGDSCTVINSKQPTIDWVMMPEA<br>AHLAEVRSYCYLATVLSQAKAAPTNGEAHDKRADPAFVRCQGVDRIG<br>WGNGCGLFGKGSIDTCAIFAKSTKATGRTLKENIKYEVAIFVHPPTTVE<br>SHGWSYQVGTQAGHFSTTPAAPSITLKLGEYGVTVQCEPSGSDITDIA<br>YYWITVGTCTFLVHREAFVQDLNLPWSAGSTVWIRETLMEFEERPHATIQ<br>SVZIALGSGEGALHQAAGAPVEFSNITVKLTSHQKCKVQKEXLQIKGT<br>TYGRLVYVNPFSVATATIAKVLIELEPPFGDSYGVVGEQZDIHHAHKS<br>GSSGKAFTTTLKGAQRALAGDTANDFGSGGVFTSVGKAVHNVFGGAF<br>RSLFGGNSKITQGLLGALLLWNGIARDRSIALTFLAVGGVLLFSVNH<br>A |

Prediction method: IEDB recommended 2.22 | Low adjusted\_rank = good binders

[Download result](#) 

Citations

Check to expand the result: ☒

| Allele         | # | Start | End | Length | Method used                   | Peptide         | Percentile Rank | Adjusted rank | Combib. core | Combib. score | Combib. percentile rank | Combib. adjusted rank | SMM align core | SMM align IC50 (nM) | SMM align percentile rank | SMM align adjusted rank | NN align core | NN align IC50 (nM) | NN align percentile rank | NN align adjusted rank |
|----------------|---|-------|-----|--------|-------------------------------|-----------------|-----------------|---------------|--------------|---------------|-------------------------|-----------------------|----------------|---------------------|---------------------------|-------------------------|---------------|--------------------|--------------------------|------------------------|
| HLA-DRB1*01:01 | 1 | 482   | 496 | 15     | Consensus (comb lib 1s/mm/nn) | ALTFLAVGGVLLFLS | 0.10            | 0.10          | FLAVGGVLL    | 0.01          | 0.01                    | 0.01                  | FLAVGGVLL      | 3.00                | 0.10                      | 0.10                    | FLAVGGVLL     | 21.10              | 7.10                     | 7.10                   |
| HLA-DRB1*01:01 | 1 | 481   | 495 | 15     | Consensus (comb lib 1s/mm/nn) | IALTFLAVGGVLLFL | 0.10            | 0.10          | FLAVGGVLL    | 0.01          | 0.01                    | 0.01                  | FLAVGGVLL      | 3.00                | 0.10                      | 0.10                    | FLAVGGVLL     | 19.10              | 6.20                     | 6.20                   |
| HLA-DRB1*01:01 | 1 | 483   | 497 | 15     | Consensus (comb lib 1s/mm/nn) | LTLAVGGVLLFLSV  | 0.10            | 0.10          | FLAVGGVLL    | 0.01          | 0.01                    | 0.01                  | FLAVGGVLL      | 3.00                | 0.10                      | 0.10                    | FLAVGGVLL     | 37.30              | 13.00                    | 13.00                  |
| HLA-DRB1*01:01 | 1 | 479   | 493 | 15     | Consensus (comb lib 1s/mm/nn) | RSIALTFLAVGGVLL | 0.10            | 0.10          | FLAVGGVLL    | 0.01          | 0.01                    | 0.01                  | LTLAVGGV       | 3.00                | 0.10                      | 0.10                    | FLAVGGVLL     | 20.10              | 6.60                     | 6.60                   |
| HLA-DRB1*01:01 | 1 | 480   | 494 | 15     | Consensus (comb lib 1s/mm/nn) | SIALTFLAVGGVLLF | 0.10            | 0.10          | FLAVGGVLL    | 0.01          | 0.01                    | 0.01                  | FLAVGGVLL      | 3.00                | 0.10                      | 0.10                    | FLAVGGVLL     | 18.60              | 6.10                     | 6.10                   |
| HLA-DRB1*01:01 | 1 | 485   | 499 | 15     | Consensus (comb lib 1s/mm/nn) | FLAVGGVLLFLSVNV | 0.91            | 0.91          | FLAVGGVLL    | 0.01          | 0.01                    | 0.01                  | FLAVGGVLL      | 9.00                | 0.91                      | 0.91                    | FLAVGGVLL     | 135.80             | 31.00                    | 31.00                  |

## IEDB Analysis Resource

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### MHC-I Binding Prediction Results

Input Sequences

| # | Name                       | Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | LCMV Armstrong, Protein GP | MGQIVTMFEALPHIIDVINIVIVITGKAVYNFATCGIFALISFLLLAGRSCGMYGLKGPDIYK<br>GVYQFQSVFEDMSHLNLTMPNACSANNSHHYISMGTSGLELTFTNDSIISHNFCNLTSAFNKK<br>TFDHTLMSIVSSSLHSIRGNSNYKAVSCDFNNGITQYNLTFSDAQSAGSQCRTRFRGRVDMF<br>RTAFGGKYMRSWGWTGSDGKTTWCSTQSYQYLIQNRWENHCTYAGPFGMSRILLSQE<br>KTKFFTRRLAGTFTWTLSOSSGVENPGGYCLTKWMILAAELKCFGNTAVAKCNVNHDAEFC<br>DMLRLIDYNKAALSKFKEDVESALHLFKTTVNSLISDQLLMRNHLRDLMGVPYCNYSKFWYL<br>EHAKTGETSVPKWLVTNGSYLNETHFSQIEQEAADNMITEMLRKDYIKRQGSTPLALMDLL<br>MFSTSAYLVSIFLHLVKIPTRHIKGGSCPKPHRLTNKGICSCGAFKVPGVKTVWKRR                                                                        |
| 2 | LCMV Armstrong, Protein NP | MSLSKEVKSQWQTALRRELQSFTSDVKAIVKADATNLLNGLDFSEVSNVQIRMRKEKRDOK<br>DLQRLRSLNQTVHSLVDLKSTSKKNVLKVGRLSAEELMSLAADLEKLKAKIMRSERPQASGV<br>YMGNLTTQQLDQRSQILQIVGMRKPQQGASGVVRVWDVKDSSLNNQFGTMMPSLTMACMA<br>KQSQTPLNDVVQALDGLLYTVKYPNLNDLERLKDKHPVLGVITEQSSINISGYNFSLGAA<br>VKAGAAALLDGGNIMLESILIKPSNSEDLLKAVLGAKRKLNMVFSDQVGDNRPNYENILYKVCSSG<br>EGWPYIACRTSIVGRAVENTTIDLTSKPAVNSPRPAPGAAGPPQVGLSYSQTMLLKDLIMGG<br>IDPNAPTWIDIEGRFNDPVEIAIFQPNQGFHIFREPVDAQKQKQDSKYSHGMDLADLFNA<br>QPGLTSSVIGALPQGMVLSQGSDDIRKLLDSQNRKDIKLDVEMTREASREYEDKYWDKYG<br>VLCKMHTGIVRDKKKKEITPHCALMDCIIFESASKARLPDLKTVHNILPHDLIRFGPNVVTL |

NetMHCpan allele distance 

| Input Allele | Closest Allele | Distance |
|--------------|----------------|----------|
| HLA-A*01:01  | HLA-A*01:01    | 0.000    |

Prediction method: NetMHCpan EL 4.1 | High Score = good binder

\* The 'IEDB recommended' method was updated in September 2020 to NetMHCpan EL 4.1 More information is available on the [help](#) page.

[Download result](#) 

Citations

| Allele      | # | Start | End | Length | Peptide  | Core      | Icore    | Score    | Percentile Rank |
|-------------|---|-------|-----|--------|----------|-----------|----------|----------|-----------------|
| HLA-A*01:01 | 2 | 198   | 205 | 8      | LTDLGLLY | LTD-LGLLY | LTDLGLLY | 0.970395 | 0.01            |
| HLA-A*01:01 | 1 | 220   | 227 | 8      | CSQTSYQY | CSQ-TSYQY | CSQTSYQY | 0.18009  | 0.59            |
| HLA-A*01:01 | 1 | 93    | 100 | 8      | SANNSHHY | SAN-NSHHY | SANNSHHY | 0.161248 | 0.64            |

## Drawbacks of IEDB

- The IEDB has certain constraints and limitations. The IEDB does not curate HIV or cancer references, as these data are not within the programmatic scope of the NIAID. However, exceptions for the inclusion of HIV and cancer epitopes are made if they are reported alongside other epitopes. Similarly, any epitope reported in the context of a cancer resulting from viral infection (e.g., human papilloma virus and human T-lymphotropic virus) are included. The database is also limited by the information provided in each paper by the authors. For example, the type of disease (disease state) is captured in accordance with the patient histories provided by the author. The IEDB primarily describes disease states using the Disease Ontology (DO), which uses formal nomenclature and synonyms. The progress of the disease (disease stage) is described using a searchable list of terms, such as acute, chronic, and post.